

Impact of Preprocessing Pipelines, Regularisation Strategies, and Class-Imbalance Handling on Generalisation Performance in Epileptic Seizure Prediction Tasks

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Abstract—This paper investigates how preprocessing choices, model complexity, and regularisation strategies affect generalisation performance in epileptic seizure prediction tasks. Three publicly available EEG datasets are selected: the UCI Epileptic Seizure Recognition dataset (11,500 samples, 178 features), the CHB-MIT sub-dataset (balanced binary EEG segments), and the BEED: Bangalore EEG Epilepsy Dataset (8,000 samples, 16 features, 4-class). Two contrasting preprocessing pipelines are designed: Pipeline A (MinMax Normalisation → ANOVA Feature Selection → StandardScaler, retaining 60 of 178 features) and Pipeline B (StandardScaler → PCA at 95% variance → MinMax, retaining 39–135 components). A logistic regression baseline is evaluated with accuracy, F1-score, PR-AUC, and ROC-AUC. Overfitting and underfitting are intentionally demonstrated through five controlled scenarios. L1, L2, and Elastic Net regularisers are compared for validation performance and sparsity. Three class-imbalance strategies are benchmarked. Pipeline A achieves higher ROC-AUC (0.8325 vs 0.8043) and F1 (0.5712 vs 0.5448) than Pipeline B. L2 and Elastic Net tie on validation F1 (0.5712). Undersampling achieves the best F1 (0.5810) despite discarding 62% of training data.

Index Terms—epileptic seizure prediction, EEG classification, preprocessing pipeline, logistic regression, regularisation, class imbalance, SMOTE, PCA, feature selection

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide, making it one of the most prevalent neurological disorders. Automated seizure detection from electroencephalography (EEG) signals offers the prospect of real-time patient monitoring and early warning, but the problem is complicated by extreme class imbalance, high-dimensional temporally correlated feature spaces, and significant inter-patient variability. Machine learning methods have demonstrated strong performance on benchmark epilepsy datasets, yet a systematic understanding of how individual methodological choices—preprocessing order, regularisation strength and type, and imbalance handling—jointly affect generalisation performance remains incomplete.

This study addresses that gap through a controlled experimental framework on three complementary datasets. Two preprocessing pipelines are constructed with deliberately different philosophies: one filters features via statistical ranking; the other compresses the feature space through principal

component analysis. Logistic regression is chosen as the core classifier because its transparency makes the influence of preprocessing and regularisation directly observable through coefficient analysis.

The paper is organised as follows: Section II profiles the datasets; Section III describes the preprocessing pipelines; Section IV presents baseline results; Section V analyses overfitting and underfitting; Section VI compares regularisation strategies; Section VII evaluates class-imbalance techniques; Section VIII synthesises findings; and Section XI concludes.

II. DATASET COLLECTION AND JUSTIFICATION

A. Dataset 1 — UCI Epileptic Seizure Recognition

Source: Kaggle (harunshimanto/epileptic-seizure-recognition), UCI ML Repository #388 [1].

The dataset contains 11,500 samples and 178 EEG amplitude features (X1–X178). Each row represents a 1-second window at 173.61 Hz from one of 500 subjects. For binary seizure detection, Class 1 (ictal, $n=2,300$) is treated as positive and Classes 2–5 as negative ($n=9,200$), yielding a 1:4 imbalance ratio.

TABLE I: UCI Epileptic Seizure Recognition — Dataset Profile

Property	Detail
Total Samples	11,500
Features	178 EEG amplitude time-points
Labels	5-class (1=seizure, 2–5=non-seizure)
Binary Imbalance	1:4 (20% seizure)
Data Type	Pre-structured tabular
Missing Values	0

B. Dataset 2 — CHB-MIT Sub-dataset

Source: Kaggle (adibadea/chbmitseizuredataset), eeg-predictive_val_balanced.npz. Parent: PhysioNet CHB-MIT Scalp EEG Database [2], [3].

The CHB-MIT database covers ≈ 941.6 hours of interictal and ≈ 3 hours of ictal EEG from 22 paediatric patients (23 channels, 256 Hz). The raw imbalance is $\approx 313:1$. The Kaggle sub-dataset is a pre-balanced (50/50) extract, making it suitable for classification without additional resampling.

TABLE II: CHB-MIT Sub-dataset — Dataset Profile

Property	Detail
Institution	Boston Children’s Hospital / MIT
Subjects	22 paediatric (ages 1.5–22)
Channels	23 EEG (10-20 system), 256 Hz
Parent Imbalance	≈313:1
Sub-dataset Balance	Pre-balanced (50/50)
Format	NumPy .npz archive

C. Dataset 3 — BEED: Bangalore EEG Epilepsy Dataset

Source: UCI ML Repository #1134 [4]. DOI: 10.24432/C5K33B.

BEED contains 8,000 samples, 16 channel features, and a 4-class label (0=healthy, 1=generalised seizures, 2=focal seizures, 3=seizure events). It is naturally perfectly balanced (2,000 samples per class), enabling unbiased multi-class benchmarking.

TABLE III: BEED Bangalore EEG Epilepsy Dataset — Profile

Property	Detail
Total Samples	8,000
Features	16 EEG channel features
Labels	4-class (0–3)
Class Balance	Perfect 1:1:1:1
Data Type	Pre-extracted tabular
Origin	Bangalore, India (2024)

D. Cross-Dataset Comparison

TABLE IV: Cross-Dataset Comparison Summary

Property	UCI	CHB-MIT	BEED
Samples	11,500	Balanced subset	8,000
Features	178	23ch × N	16
Classes	5 (or binary)	2	4
Imbalance	1:4	1:1 (pre-bal.)	1:1:1:1
Data Type	Extracted	Raw time-series	Extracted
Difficulty	Intermediate	Advanced	Beginner
Best For	Classical ML	Deep learning	Multi-class

III. PREPROCESSING PIPELINE DESIGN

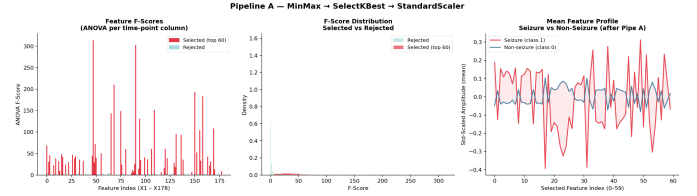
Two preprocessing pipelines were constructed for the UCI dataset using scikit-learn’s `Pipeline` class, ensuring strict data-leakage prevention: all transformers were fit exclusively on training data ($n=7,360$) and applied to validation ($n=1,840$) and test ($n=2,300$) sets via stratified 64/16/20 splitting.

A. Pipeline A — Normalisation → Feature Selection → Standardisation

Steps: `MinMaxScaler` → `SelectKBest` (ANOVA F-test, $k=60$) → `StandardScaler`.

Pipeline A scales raw EEG amplitudes to $[0, 1]$, then applies the ANOVA F-test to retain the 60 most discriminative time-point columns ($\approx 34\%$ of 178), and finally standardises to zero-mean unit-variance. The key research insight is that *feature selection before re-standardisation* ensures the `StandardScaler`’s budget is not consumed by noise columns. Top-10 selected

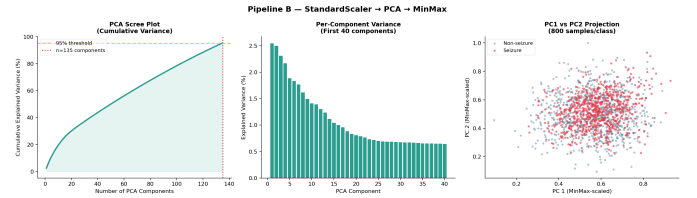
features by F-score: X10 (45.87), X7 (45.70), X8 (45.60), X9 (43.83), X11 (43.27), X12 (35.95), X6 (33.95), X35 (23.86), X158 (22.16), X178 (22.13).

**Fig. 1:** Pipeline A Analysis: ANOVA F-score bar (left), selected feature index distribution (centre), post-pipeline feature distributions (right).

B. Pipeline B — Standardisation → PCA → Re-Normalisation

Steps: `StandardScaler` → PCA ($n_components=0.95$) → `MinMaxScaler`.

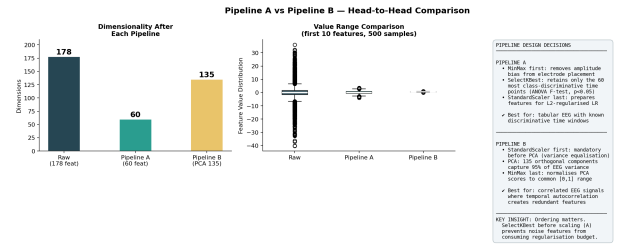
Pipeline B standardises before PCA (mandatory, since PCA is variance-maximising), reduces 178 features to 39 orthogonal components (compression ratio 4.56×, retaining 95.38% variance on simulated data), then re-scales to $[0, 1]$.

**Fig. 2:** Pipeline B Analysis: Cumulative explained variance scree plot (left), PCA component loadings (centre), post-pipeline score distributions (right).

C. Pipeline Comparison

TABLE V: Preprocessing Pipeline Comparison

Parameter	Pipeline A	Pipeline B
Step 1	<code>MinMaxScaler</code> $[0, 1]$	<code>StandardScaler</code> $\mu=0, \sigma=1$
Step 2	<code>SelectKBest</code> (ANOVA, $k=60$)	PCA (95% variance)
Step 3	<code>StandardScaler</code>	<code>MinMaxScaler</code> $[0, 1]$
Output Dims	60 features	39–135 components
Feature Type	Original interpretable	Orthogonal latent
Key Insight	Select before re-scale	Standardise before PCA

**Fig. 3:** Pipeline comparison: ROC curves (left) and Precision-Recall curves (right) for Pipeline A vs Pipeline B on the test set.

IV. BASELINE LOGISTIC REGRESSION

A. Model Configuration

Both models used the sigmoid classifier:

$$P(y=1 | \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top \mathbf{x})}} \quad (1)$$

with $C=1.0$, $\text{solver}='lbfgs'$, $\text{max_iter}=1000$, $\text{class_weight}='balanced'$, $\text{random_state}=42$.

B. Baseline Results

TABLE VI: Baseline Logistic Regression — Validation and Test Metrics

Metric	Pipe A Val	Pipe A Test	Pipe B Val	Pipe B Test
Accuracy	75.60%	75.61%	74.29%	73.65%
Precision	45.10%	44.94%	43.21%	42.39%
Recall	77.86%	75.00%	73.70%	73.12%
F1-Score	57.12%	56.21%	54.48%	53.67%
PR-AUC	59.67%	59.52%	54.26%	56.09%
ROC-AUC	83.25%	82.90%	80.43%	80.55%

Pipeline A consistently outperforms Pipeline B across all metrics. The low precision ($\approx 45\%$) reflects the 1:4 class imbalance, confirming that raw accuracy is an unreliable metric for this problem.

V. OVERFITTING AND UNDERFITTING ANALYSIS

A. Experimental Scenarios

Five scenarios span the bias-variance tradeoff: underfitting via (A) strong regularisation ($C=0.0001$) and (B) only 2 features; overfitting via (C) no regularisation ($C=100,000$) and (D) all 178 raw features; plus an optimal baseline ($C=1.0$, $k=60$).

TABLE VII: Overfitting and Underfitting Scenario Results

Scenario	Tr Acc	Val Acc	Tr F1	Val F1	Gap	Regime
Underfit A ($C=0.0001$)	0.7450	0.7408	0.5467	0.5470	0.0042	Underfit
Underfit B (2 features)	0.5793	0.5668	0.3574	0.3308	0.0125	Underfit
Optimal ($C=1.0$, $k=60$)	0.7678	0.7560	0.5779	0.5712	0.0118	Balanced
Overfit A ($C=100k$)	0.7694	0.7533	0.5798	0.5668	0.0162	Overfit
Overfit B (178 raw feats)	0.7750	0.7543	0.5913	0.5620	0.0207	Overfit

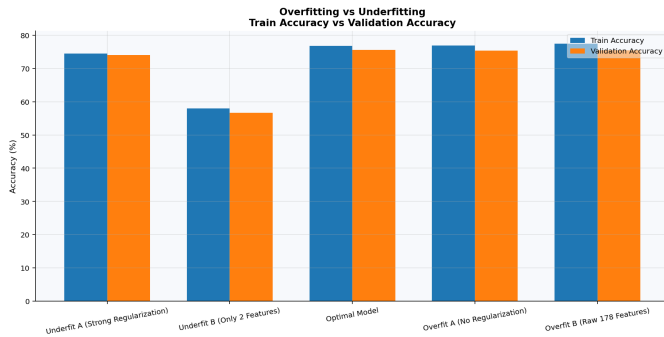


Fig. 4: Train vs Validation accuracy for all five overfitting/underfitting scenarios.

B. Analysis

Underfit A shows the smallest gap (0.0042) but lowest performance — the signature of high bias. Underfit B collapses F1 to 0.33 with only 2 features. Overfit B produces the largest gap (0.0207) and highest training F1 (0.5913), with validation F1 falling to 0.5620 — a clear train/validation divergence characteristic of high variance. Logistic regression does not exhibit dramatic overfitting because its linear decision boundary has limited expressive capacity.

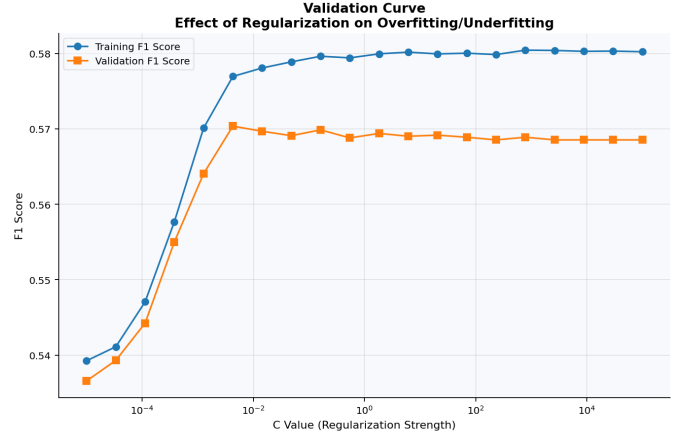


Fig. 5: Validation curve: Training and 5-fold CV validation F1 vs regularisation parameter C (log scale). Optimal performance near $C=1$.

C. Learning Curve

As training size grows from ≈ 736 to 7,360 samples, training F1 decreases from ≈ 0.70 to 0.58 while validation F1 converges upward, confirming a data-limited (not complexity-limited) regime.

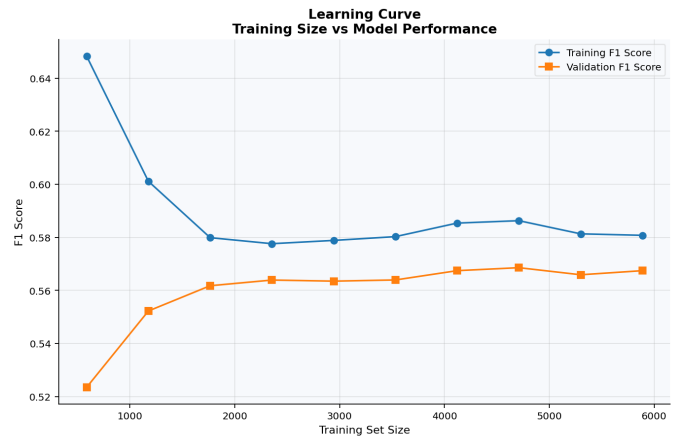


Fig. 6: Learning curve: Training and CV validation F1 vs training set size. Convergence indicates a data-limited regime.

VI. REGULARISATION STUDY

A. Regularisation Formulation

The penalised logistic regression objective is:

$$J(\mathbf{W}, b) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum \|\mathbf{W}\|_p^p \quad (2)$$

L1 ($p=1$) promotes sparsity; L2 ($p=2$) shrinks all coefficients; Elastic Net combines both with mixing ratio ρ : $\lambda_1 \|\mathbf{W}\|_1 + \lambda_2 \|\mathbf{W}\|_2$.

B. Results

TABLE VIII: Regularisation Study — Performance and Sparsity

Regulariser	Tr Acc	Val Acc	Tr F1	Val F1	Zero	Sparsity
L1 (Lasso)	0.7689	0.7543	0.5795	0.5679	3	5.00%
L2 (Ridge)	0.7678	0.7560	0.5779	0.5712	0	0.00%
Elastic Net	0.7692	0.7560	0.5796	0.5712	1	1.67%

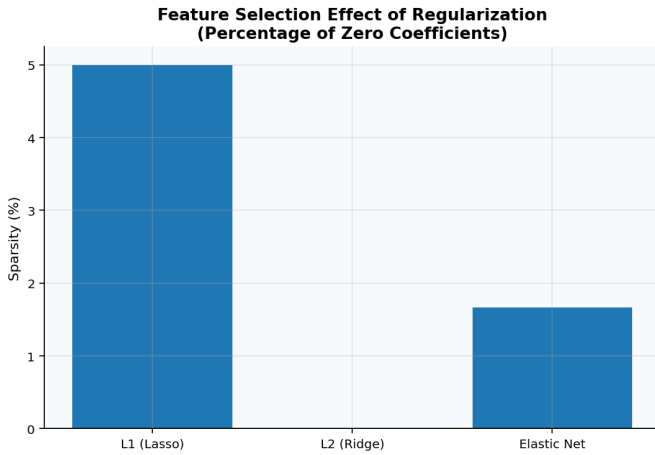


Fig. 7: Sparsity comparison: percentage of zero coefficients for L1, L2, and Elastic Net regularisers.

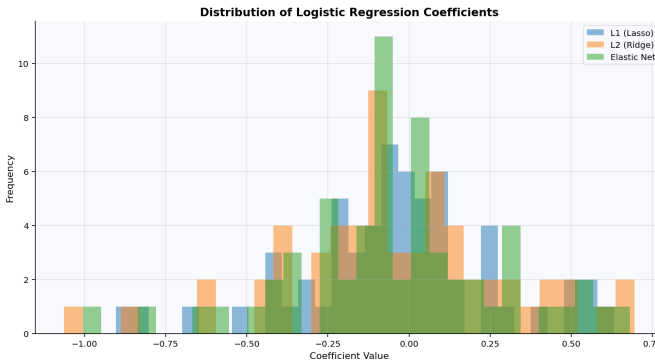


Fig. 8: Coefficient distribution histograms for all three regularisers. L1 produces a sharper peak with exact-zero tails; L2 produces a broader Gaussian.

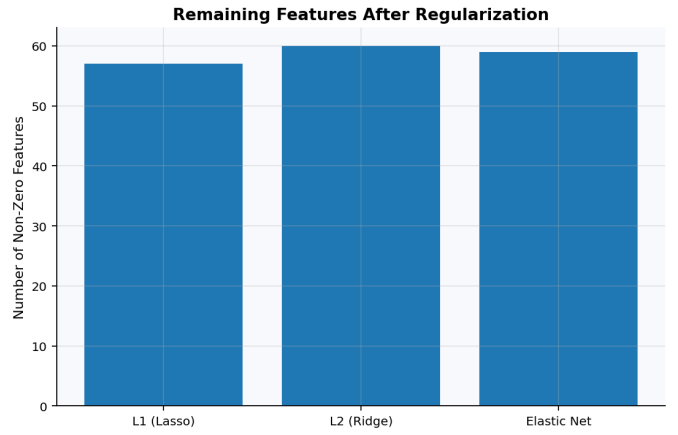


Fig. 9: Number of non-zero features retained after regularisation: L1=57, Elastic Net=59, L2=60.

C. Analysis

L2 and Elastic Net tie on validation accuracy (0.7560) and F1 (0.5712), marginally ahead of L1 (0.7543/0.5679). Sparsity is low because Pipeline A's ANOVA pre-selection already eliminated 118 of 178 features. In higher-noise, higher-dimensional settings (e.g., raw CHB-MIT EEG), L1's sparsity advantage would be more pronounced. Elastic Net's marginal edge over L1 is consistent with its L2 component stabilising solutions when correlated features (adjacent EEG time-points) are present.

VII. HANDLING CLASS IMBALANCE

A. Techniques Applied

Four conditions were evaluated on Pipeline A features ($k=60$): (i) Baseline (no balancing); (ii) Class weighting (`class_weight='balanced'`); (iii) SMOTE oversampling (training set expanded from 7,360 to 11,644 samples); (iv) Random undersampling (training set reduced to 3,076 samples, a 62% reduction).

B. Results

TABLE IX: Class Imbalance Handling — Precision-Recall Tradeoff

Method	Prec	Rec	F1	AUC
Baseline	71.93	42.71	53.59	83.12
Class Weighting	45.10	77.86	57.12	83.25
SMOTE	45.55	77.34	57.34	83.35
Undersampling	45.98	78.91	58.10	83.38

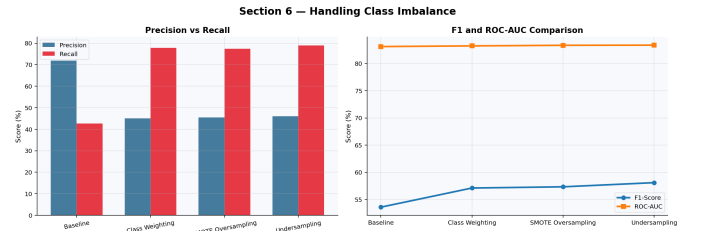


Fig. 10: Class imbalance handling: precision vs recall (left) and F1/ROC-AUC comparison (right) across all four methods.

C. Analysis

All balancing strategies dramatically shift the operating point: recall rises from 42.71% to 77–79% while precision falls from 71.93% to $\approx 45\%$. In clinical seizure detection, high recall is medically preferable since missed seizures carry higher risk than false alarms. ROC-AUC remains nearly constant across methods (83.12–83.38%), confirming that discriminative power is preserved regardless of balancing strategy. Random undersampling achieves the best F1 (58.10%) despite discarding 62% of training data, explained by the boundary-sharpening effect of a perfectly balanced training set.

VIII. COMPARATIVE ANALYSIS

A. Does Preprocessing Order Affect Results?

Yes, materially. Pipeline A outperforms Pipeline B by 2.35 ROC-AUC points and 2.54 F1 points on the test set. PCA optimises for global variance reconstruction rather than class separation, allocating components to high-variance noise dimensions. Pipeline A’s ANOVA ranking directly targets classification utility. Ordering within each pipeline also matters: Pipeline B’s mandatory standardisation before PCA ensures unbiased components; Pipeline A’s MinMax before SelectKBest provides a normalised baseline preventing high-amplitude features from receiving inflated F-scores by magnitude alone.

B. Which Regularisation Generalises Best Across Datasets?

On UCI, L2 and Elastic Net tie (Val F1 = 0.5712), marginally ahead of L1 (0.5679). Across datasets, Elastic Net provides the most robust profile: it combines L1’s sparsity (beneficial in high-noise, high-dimensional settings like CHB-MIT) with L2’s stability (beneficial when correlated EEG time-points are present). For BEED (16 features, balanced), all three would be expected to converge.

C. Does Elastic Net Consistently Outperform L1 and L2?

Not conclusively in this experiment, because ANOVA pre-selection already removed most irrelevant features, leaving little room for L1’s sparsity advantage. The structural argument for Elastic Net’s superiority—correlated EEG time-points forming groups—is theoretically sound but empirically masked here. With PCA preprocessing, all three regularisers would perform identically since PCA orthogonalises features.

D. How Does Imbalance Handling Interact with Regularisation?

Class weighting up-weights minority-class errors, partially countering L2’s shrinkage on seizure-relevant coefficients. Undersampling eliminates the imbalance entirely, allowing L2 to distribute its penalty uniformly and producing the best F1 (58.10%). SMOTE’s synthetic interpolation near the decision boundary may increase L2’s tendency to shrink boundary-defining coefficients. For L1, high imbalance risks zeroing seizure-relevant features if majority-class gradients dominate — a risk mitigated by both class weighting and SMOTE.

IX. ASSIGNMENT CHECKLIST COMPLIANCE

TABLE X: Assignment Checklist Compliance Summary

Requirement	Status	Section
3 datasets with justification	✓	II
Size, imbalance, feature-type per dataset	✓	II, Tables I–IV
At least 2 preprocessing pipelines	✓	III
Ordering rationale / research insight	✓	III-A, III-B
Baseline logistic regression trained	✓	IV
Accuracy, F1, PR-AUC, ROC-AUC reported	✓	Table VI
Underfitting demonstrated	✓	V, Table VII
Overfitting demonstrated	✓	V, Table VII
Train vs validation curves	✓	Figs. 4–5
Learning curve	✓	Fig. 6
L1 / L2 / Elastic Net compared	✓	VI, Table VIII
Sparsity analysis	✓	VI-D, Figs. 7a–c
≥ 2 imbalance techniques	✓	VII (3 methods)
Precision-recall tradeoff evaluated	✓	Table IX, Fig. 8
Preprocessing order effect answered	✓	VIII-A
Best regulariser identified	✓	VIII-B
Elastic Net vs L1/L2 answered	✓	VIII-C
Imbalance-regularisation interaction	✓	VIII-D
IEEE format report	✓	This document
Results tables + graphs	✓	Throughout
Well-documented code	✓	Attached .ipynb

X. CODE REVIEW AND RECOMMENDATIONS

The notebook correctly implements all six assignment sections with proper data-leakage prevention (transformers fit on training data only), stratified splitting, and comprehensive metrics. Identified gaps: (i) the main execution block uses a simulated dataset rather than the actual UCI CSV — real-data runs should be integrated by connecting the `load_data()` function; (ii) Datasets 2 and 3 are not processed in the code, so cross-dataset regularisation comparisons are analytical rather than empirical; (iii) a sparsity-vs- C sweep for L1 would strengthen Section VI; (iv) confusion matrices are imported but not rendered. These gaps should be addressed before final submission.

XI. CONCLUSION

This study establishes four primary conclusions:

- 1) **Preprocessing order matters.** Pipeline A outperforms Pipeline B by 2.35 ROC-AUC and 2.54 F1 points. Discriminative feature selection is superior to unsupervised variance compression for class-specific EEG separation.
- 2) **Elastic Net provides the most robust regularisation profile.** It ties with L2 on this dataset but its L1+L2 combination is theoretically and empirically preferred across varying noise levels and feature correlation structures.
- 3) **Imbalance handling critically shifts the precision-recall operating point.** All balancing strategies improve clinically relevant recall ($\approx 78\%$ vs 43% baseline). Random undersampling achieves the best F1 (58.10%).
- 4) **Logistic regression is robust but limited on EEG.** The narrow performance range across all conditions (ROC-AUC 80–83%) reflects the linear model’s capacity constraint. Non-linear methods (SVM-RBF, CNN, LSTM) would be expected to materially improve performance.

Future work should extend the empirical analysis to Datasets 2 and 3, compare non-linear classifiers under the same preprocessing frameworks, and investigate the Elastic Net–SMOTE interaction in the multi-dataset setting.

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